

INTERNSHIP REPORT– TASK 3

Title : Decision Classifier for Customer Purchase Prediction



Student Details

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- **Internship Domain:** Data Science

1. Objective

The objective of this task is to build a **Decision Tree Classification model** to predict whether a customer will subscribe to a product or service based on demographic and behavioral data.

2. Dataset Used

The dataset used is the **Bank Marketing Dataset**, which contains information about customers contacted by a bank.

Features include:

- Age
- Job
- Marital status
- Education
- Balance
- Contact type
- Duration
- Campaign details

Target variable:

- $y \rightarrow$ Whether the customer subscribed (Yes/No)

3. Tools & Technologies Used

- Python
- Google Colab
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn

4. Data Preprocessing

- Handled categorical variables using **OneHotEncoding**
- Split dataset into training and testing sets (80:20)
- Ensured clean numerical input for model training

5. Model Used

- Algorithm: **Decision Tree Classifier**
- Criterion: Entropy
- Max Depth: 5

6. Model Training Process

- The dataset was split into training and testing data.
- Decision Tree model was trained using training data.
- Model learned patterns between customer attributes and subscription behavior.

7. Model Evaluation

The model was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

8. Visualization

A Decision Tree was plotted to understand how decisions are made based on features.

9. Key Learnings

- Understanding classification problems
- Working with categorical data
- Decision Tree algorithm behavior
- Importance of preprocessing in ML
- Model evaluation techniques

10. Conclusion

The Decision Tree model successfully predicts whether a customer will subscribe to a product/service. This helps businesses improve targeted marketing strategies and customer conversion.